

WASP-36b

R-0769

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_250101 · created 2026-07-28T05:00:40+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460676.9585 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.047
Transit depth (Rp/Rs)^2	2.76 +/- 1.58 [%]
Orbital Inclination (inc)	82.46 +/- 2.86
Airmass coefficient 1 (a1)	0.167 +/- 0.09
Airmass coefficient 2 (a2)	1.18 +/- 0.67
Scatter in the residuals of the lightcurve fit is	49.4 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.065 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.047 vs 0.1368 (deviation 0.59σ)
Duration fit vs published	0.065 d vs 0.0758 d (deviation 0.38σ)
Mid-time O–C	-11.4 min
Depth SNR	3.3
Residual scatter	49.4 %

Light curve

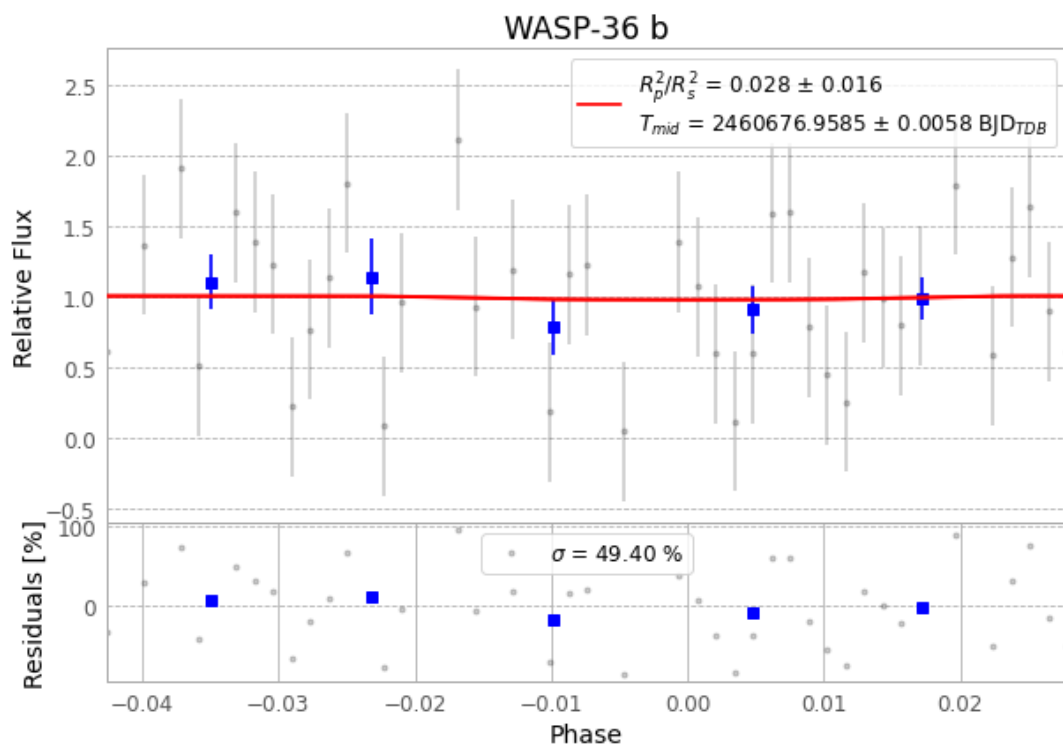


Figure 1 — FinalLightCurve_WASP-36 b_2025-01-01.png (SHA-256 02ae5d2d07d5846b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [599, 417], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53fd75422c3e4a5e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

53 FITS frames (35 MB), WASP-36250101091815.FITS → WASP-36250101115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-250101.log (d505f0300d8d...), FinalLightCurve_WASP-36 b_2025-01-01.png (02ae5d2d07d5...), FinalLightCurve_WASP-36 b_2025-01-01.csv (c1b4744d49f9...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:00Z.